

QiskitError: 'Sum of amplitudes-squared does not equal one.' in QEOM wh

<https://github.com/qiskit-community/qiskit-nature/issues/373>

ROW 84

QiskitError: 'Sum of amplitudes-squared does not equal one.' in QEOM when two qubit reduction is set True #373

New issue

Closed

#432



HuangJunye opened on Sep 24, 2021 · edited by HuangJunye

Edits ...

Information

- Qiskit Nature version: 0.2.0
- Python version: 3.8.8
- Operating system: macOS 11.4 Big Sur

What is the current behavior?

Error message: QiskitError: 'Sum of amplitudes-squared does not equal one.'

```
-----
QiskitError                                Traceback (most recent call last)
<ipython-input-6-a9e19ed18bca> in <module>
      6 vqe_ground_state_solver = GroundStateEigensolver(qubit_converter=converter, solver=algorithm)
      7 qeom_excited_state_solver = QEOM(ground_state_solver=vqe_ground_state_solver)
----> 8 qeom_results = qeom_excited_state_solver.solve(problem=problem)
      9
     10 ##### Write your code between these lines

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit_nature/algorithms/excited_states_solvers/
110
111     # 3. Evaluate eom operators
--> 112     measurement_results = self._gsc.evaluate_operators(
113         groundstate_result.eigenstates[0], matrix_operators_dict
114     )

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit_nature/algorithms/ground_state_solvers/
164         results[name] = None
165     else:
--> 166         results[name] = self._eval_op(state, op, quantum_instance, expectation)
167     else:
168         if operators is None:

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit_nature/algorithms/ground_state_solvers/
187         if expectation is not None:
188             exp = expectation.convert(exp)
--> 189             result = sampler.convert(exp).eval()
190         except ValueError:
191             # TODO make this cleaner. The reason for it being here is that some quantum

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/opflow/converters/circuit_sampler.py in
174
175     # convert to circuit and reduce
--> 176     operator_dicts_replaced = operator.to_circuit_op()
177     self._reduced_op_cache = operator_dicts_replaced.reduce()
178

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/opflow/list_ops/list_op.py in to_circuit
558     ).reduce()
559     return self.__class__(
--> 560     [
561         op.to_circuit_op() if not isinstance(op, OperatorStateFn) else op
562         for op in self.oplist

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/opflow/list_ops/list_op.py in <listcomp>
559     return self.__class__(
560     [
--> 561         op.to_circuit_op() if not isinstance(op, OperatorStateFn) else op
562         for op in self.oplist
563     ],

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/opflow/state_fns/vector_state_fn.py in t
162     from .circuit_state_fn import CircuitStateFn
163
--> 164     csfn = CircuitStateFn.from_vector(self.primitive.data) * self.coeff
165     return csfn.adjoint() if self.is_measurement else csfn
166

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/opflow/state_fns/circuit_state_fn.py in
124     normalization_coeff = np.linalg.norm(statevector)
125     normalized_sv = statevector / normalization_coeff
--> 126     return CircuitStateFn(initialize(normalized_sv), coeff=normalization_coeff)
```

Assignees

mrossinek

woodsp-ibm

Labels

status: confirmed type: bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix EOM with 2Q reduction and tapering
qiskit-community/qiskit-nature

Notifications

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Participants



```

--> 126         return CircuitStateFn(Initialize(normalized_sv), coeff=normalization_coeff)
127
128     def primitive_strings(self) -> Set[str]:

/usr/local/anaconda3/envs/oled-new/lib/python3.8/site-packages/qiskit/extensions/quantum_initializer/initializ
90     # Check if probabilities (amplitudes squared) sum to 1
91     if not math.isclose(sum(np.absolute(params) ** 2), 1.0, abs_tol=EPS):
--> 92         raise QiskitError("Sum of amplitudes-squared does not equal one.")
93
94     num_qubits = int(num_qubits)

QiskitError: 'Sum of amplitudes-squared does not equal one.'

```

Steps to reproduce the problem

Add these code after the [example code in the README](#).

```

from qiskit_nature.algorithms import QEOM
from qiskit_nature.algorithms import GroundStateEigensolver

vqe_ground_state_solver = GroundStateEigensolver(qubit_converter=converter, solver=algorithm)
qeom_excited_state_solver = QEOM(ground_state_solver=vqe_ground_state_solver)
qeom_results = qeom_excited_state_solver.solve(problem=problem)

print(qeom_results)

```

What is the expected behavior?

Returns ElectronicStructureResult. This is the result when `two_qubit_reduction` is set to `False` in `QubitConverter`:

```

=== GROUND STATE ENERGY ===

* Electronic ground state energy (Hartree): -1.857107039335
- computed part: -1.857107039335
~ Nuclear repulsion energy (Hartree): 0.719968994449
> Total ground state energy (Hartree): -1.137138044886

=== EXCITED STATE ENERGIES ===

1:
* Electronic excited state energy (Hartree): -1.244589346215
> Total excited state energy (Hartree): -0.524620351766
2:
* Electronic excited state energy (Hartree): -0.882883155693
> Total excited state energy (Hartree): -0.162914161244
3:
* Electronic excited state energy (Hartree): -0.225098418795
> Total excited state energy (Hartree): 0.494870575654

=== MEASURED OBSERVABLES ===

0: # Particles: 2.000 S: 0.000 S^2: 0.000 M: 0.000

=== DIPOLE MOMENTS ===

~ Nuclear dipole moment (a.u.): [0.0 0.0 1.3889487]

0:
* Electronic dipole moment (a.u.): [0.0 0.0 1.40580143]
- computed part: [0.0 0.0 1.40580143]
> Dipole moment (a.u.): [0.0 0.0 -0.01685273] Total: 0.01685273
(debye): [0.0 0.0 -0.04283535] Total: 0.04283535

```

Suggested solutions



HuangJunye on Sep 24, 2021

Contributor

Author

...

Here's the gist of a notebook reproducing the error:

<https://gist.github.com/HuangJunye/cce1a2c2f6a04cc865eeb54170288acb>



Goodnot

 mrossinek added status: confirmed type: bug on Sep 29, 2021

 mrossinek on Sep 29, 2021

Member ...

This issue boils down to a mismatching qubit operator of the ground state Hamiltonian and the hopping operators. All of these are checked for their commutativity. QEOM appears to rely on using the *untapered* qubit op [here](#). However, the hopping operators are actually already being mapped+reduced+tapered. This causes a mismatch.

The question is now whether the reliance on using the untapered operator is actually required or whether using `convert_match` instead of `map` on the line linked above is sufficient. It appears to fix this particular bug but it will still cause problems when using `zsymmetry_reduction="auto"` (in which case a straight forward test appears to run into the same issue as [#342](#)).

If QEOM needs to perform tapering manually then the internal `Z2Symmetry` which is being used inside of `TwoQubitReduction` needs to be made available in `QubitConverter.zsymmetries` (which is where QEOM gets the symmetry information from).



 mrossinek on Sep 29, 2021 · edited by mrossinek

Edits ▾ Member ...

Here is a "partial fix" which changes `map` → `convert_match` and adds two unittests for QEOM:

- one with two-qubit reduction (which works with this applied patch)
- one with automatic `Z2Symmetries` (which runs into [EDIT: here used to be a linked issue. But it was the wrong ID and I don't recall which one was supposed to be linked to.])

```
diff --git a/qiskit_nature/algorithms/excited_states_solvers/qeom.py b/qiskit_nature/algorithms/excited_states_solvers/qeom.py
index d0987c189..8b98c0252 100644
--- a/qiskit_nature/algorithms/excited_states_solvers/qeom.py
+++ b/qiskit_nature/algorithms/excited_states_solvers/qeom.py
@@ -105,7 +105,7 @@ class QEOM(ExcitedStatesSolver):
     groundstate_result = self._gsc.solve(problem)

     # 2. Prepare the excitation operators
-    self._untapered_qubit_op_main = self._gsc._qubit_converter.map(problem.second_q_ops()[0])
+    self._untapered_qubit_op_main = self._gsc._qubit_converter.convert_match(problem.second_q_ops()[0])
     matrix_operators_dict, size = self._prepare_matrix_operators(problem)

     # 3. Evaluate eom operators
diff --git a/test/algorithms/excited_state_solvers/test_excited_states_solvers.py b/test/algorithms/excited_state_solvers/test_excited_states_solvers.py
index f1eac8f7e..34c5e859b 100644
--- a/test/algorithms/excited_state_solvers/test_excited_states_solvers.py
+++ b/test/algorithms/excited_state_solvers/test_excited_states_solvers.py
@@ -21,7 +21,7 @@ from qiskit.algorithms import NumPyMinimumEigensolver, NumPyEigensolver

 from qiskit_nature.drivers import UnitsType
 from qiskit_nature.drivers.second_quantization import PySCFDriver
-from qiskit_nature.mappers.second_quantization import JordanWignerMapper
+from qiskit_nature.mappers.second_quantization import JordanWignerMapper, ParityMapper
 from qiskit_nature.converters.second_quantization import QubitConverter
 from qiskit_nature.problems.second_quantization import ElectronicStructureProblem
 from qiskit_nature.algorithms import (
@@ -85,6 +85,28 @@ class TestNumericalQEOMESCCalculation(QiskitNatureTestCase):
     for idx, energy in enumerate(self.reference_energies):
         self.assertAlmostEqual(results.computed_energies[idx], energy, places=4)

+    def test_vqe_mes_2q(self):
+        """Test VQEUCSDFactory with QEOM and 2-qubit reduction"""
+        converter = QubitConverter(ParityMapper(), two_qubit_reduction=True)
+        solver = VQEUCSDFactory(self.quantum_instance)
+        gsc = GroundStateEigensolver(converter, solver)
+        esc = QEOM(gsc, "sd")
+        results = esc.solve(self.electronic_structure_problem)
+
+        for idx, energy in enumerate(self.reference_energies):
+            self.assertAlmostEqual(results.computed_energies[idx], energy, places=4)
```

```

@@ -105,7 +105,7 @@ class QEOM(ExcitedStatesSolver):
    groundstate_result = self._gsc.solve(problem)

    # 2. Prepare the excitation operators
-    self.untapered_qubit_op_main = self._gsc._qubit_converter.map(problem.second_q_ops()[0])
+    self.untapered_qubit_op_main = self._gsc._qubit_converter.convert_match(problem.second_q_ops()[0])
    matrix_operators_dict, size = self._prepare_matrix_operators(problem)

    # 3. Evaluate eom operators
diff --git a/test/algorithms/excited_state_solvers/test_excited_states_solvers.py b/test/algorithms/excited_st
index f1eac8f7e..34c5e859b 100644
--- a/test/algorithms/excited_state_solvers/test_excited_states_solvers.py
+++ b/test/algorithms/excited_state_solvers/test_excited_states_solvers.py
@@ -21,7 +21,7 @@ from qiskit.algorithms import NumPyMinimumEigensolver, NumPyEigensolver

from qiskit_nature.drivers import UnitsType
from qiskit_nature.drivers.second_quantization import PySCFDriver
- from qiskit_nature.mappers.second_quantization import JordanWignerMapper
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    for idx, energy in enumerate(self.reference_energies):
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+    def test_vqe_mes_2q(self):
+        """Test VQEUCSDFactory with QEOM and 2-qubit reduction"""
+        converter = QubitConverter(ParityMapper(), two_qubit_reduction=True)
+        solver = VQEUCSDFactory(self.quantum_instance)
+        gsc = GroundStateEigensolver(converter, solver)
+        esc = QEOM(gsc, "sd")
+        results = esc.solve(self.electronic_structure_problem)
+
+        for idx, energy in enumerate(self.reference_energies):
+            self.assertEqual(results.computed_energies[idx], energy, places=4)
+
+    def test_vqe_mes_z2(self):
+        """Test VQEUCSDFactory with QEOM and Z2Symmetries"""
+        converter = QubitConverter(ParityMapper(), two_qubit_reduction=True, z2symmetry_reduction="auto")
+        solver = VQEUCSDFactory(self.quantum_instance)
+        gsc = GroundStateEigensolver(converter, solver)
+        esc = QEOM(gsc, "sd")
+        results = esc.solve(self.electronic_structure_problem)
+
+        for idx, energy in enumerate(self.reference_energies):
+            self.assertEqual(results.computed_energies[idx], energy, places=4)
+
    def test_numpy_factory(self):
        """Test NumPyEigensolverFactory with ExcitedStatesEigensolver"""

```



mrossinek assigned [mrossinek](#) and [woodsp-ibm](#) on Sep 29, 2021



woodsp-ibm mentioned this on Nov 16, 2021

🔗 [Fix EOM with 2Q reduction and tapering #432](#)



mrossinek closed this as [completed](#) in [#432](#) on Nov 22, 2021

Fix EOM with 2Q reduction and tapering #432

<> Code

Merged mrossinek merged 7 commits into qiskit-community:main from woodsp-ibm:qeom_fix on Nov 22, 2021

Conversation 6 Commits 7 Checks 0 Files changed 6

+70 -10



woodsp-ibm commented on Nov 16, 2021 • edited

Member

Change so that operator, and hopping operators, are just mapped and possibly 2Q reduced (i.e. no general Z2Symmetry tapering). This matches the QEOM logic from before in Aqua. Fixed the (anti)-commutation logic and changed so it no longer uses deprecated logic (this logic was not covered by the unit tests before at all and was missed when other use of deprecated code was taken care of recently).

Fixes #373

I have added some unit tests - I chose to do it this way, rather than using ddt, as its easier to run one individually.

- ✓ Add release note
- ✓ Sort out remaining failure when Parity mapping is used without 2Q reduction but with Z2 symmetries. In looking at/fixing [QiskitError: 'Sum of amplitudes-squared does not equal one.' in QEOM when two qubit reduction is set True](#) #373 I noticed, when trying out various configurations to check the ground state that this same configuration, with the molecule in that code sample from that issue, would give an incorrect result. It did not fail but it seemed to be choosing a wrong sector (exact diagonalization via NumPy also gave same answer so it was not a failure of VQE, as used in that sample, to converge). I am not sure if the failure here when using UCCSD, is the exact same issue; but it's the same conversion config so it seems related.

Update: The above was indeed choosing the wrong sector. The sector selector based on HF bitstring was unchanged from the implementation in Aqua chemistry. However the HF bitstring was changed, when the code was refactored/moved to Nature, so that things are all little endian - and as such the HF bitstring was reversed. The pick sector in Aqua chemistry had the symmetry terms reversed sym.z[:-1] which would have aligned to the way it was needed in Aqua, since the HF string is now reversed in Nature I removed the sym.z reversal and the sector now selects the correct one.



Fix EOM with 2Q reduction and tapering

fb83d34



woodsp-ibm requested review from manaelmarques, mrossinek, pbark and stefan-woerner as code owners 4 years ago



woodsp-ibm and others added 2 commits 4 years ago



Merge branch 'main' into qeom_fix

Verified

479912e



Fix sector picker

c37655d



coveralls commented on Nov 19, 2021 • edited

...

Pull Request Test Coverage Report for Build 1490536673

- 2 of 7 (28.57%) changed or added relevant lines in 3 files are covered.
- No unchanged relevant lines lost coverage.
- Overall coverage increased (+0.07%) to 85.63%

Reviewers

	mrossinek	✓
	manaelmarques	•
	pbark	•
	stefan-woerner	•

Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[QiskitError: 'Sum of amplitudes-squared doe...](#)

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